CNG Vehicles in Delhi and Swachh Bharat Abhiyan

Cleaner fuel, cleaner streets: two public policies and the goal of a healthier city

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Delhi Metro and CNG buses, Azadpur. Photo: Varun Shiv Kapur, CC BY 2.0, Wikimedia Commons

Why Delhi needs cleaner air and cleaner streets

Air pollution

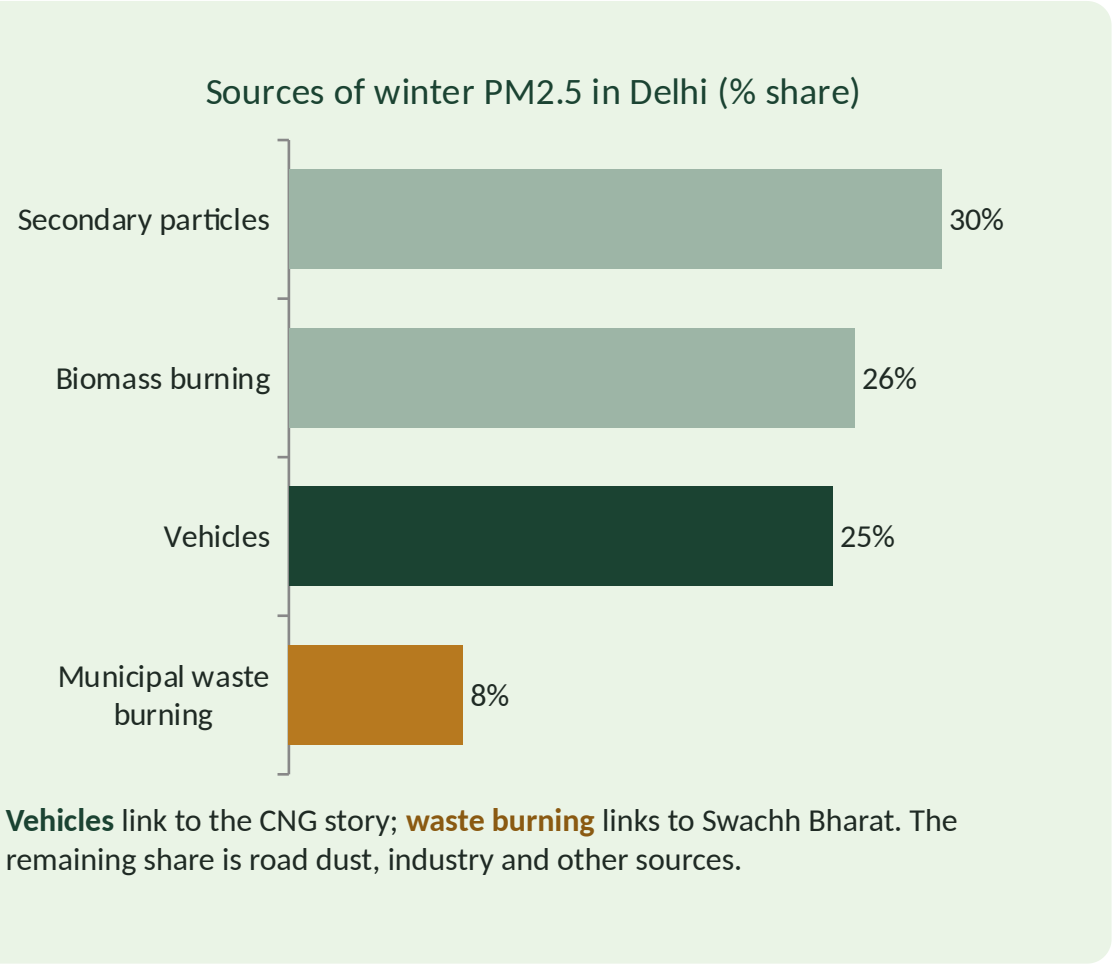
In winter, Delhi's air carries high levels of PM2.5, fine particles that reach deep into the lungs. Vehicles are one of the largest sources.

Waste and sanitation

Open dumps, burnt garbage and poor sanitation spread disease, and burning waste also adds smoke to the air.

What this presentation covers

Two separate responses: Delhi's switch to CNG (from 1998) and Swachh Bharat Abhiyan (from 2014), their results, limits, and where they meet.



Vehicles link to the CNG story; **waste burning** links to Swachh Bharat. The remaining share is road dust, industry and other sources.

Source: IIT Kanpur (2016), Comprehensive Study on Air Pollution and GHGs in Delhi, report to the Govt. of NCT of Delhi (winter PM2.5 shares, as reported by Governance Now, 7 Nov 2016).

CNG in Delhi: from a court case to the roads

What is CNG?

- Natural gas (mostly methane) squeezed to about 200–250 bar and stored in strong cylinders
- Burns in a spark-ignition engine, like petrol; diesel buses had to be replaced or re-engined
- Produces far less soot than diesel



CNG bus, Delhi. Photo: Kool Nerd 123, CC BY 4.0

1985	M.C. Mehta files a public interest petition on vehicle pollution in Delhi (WP(C) 13029/1985)
Jan 1998	EPCA (Bhure Lal Committee) set up under Section 3 of the Environment (Protection) Act, 1986
28 Jul 1998	Supreme Court accepts EPCA's plan: Delhi's whole city-bus fleet to run on CNG by 31 March 2001
2001	Deadline largely missed — too few CNG buses were ready; Court allows only conditional extensions
5 Apr 2002	Court calls CNG a clean fuel; diesel buses fined ₹500 a day, rising to ₹1,000
2021	CAQM Act replaces EPCA with a statutory Commission for Air Quality Management
2025–26	Delhi shifts to electric buses; CNG cluster buses to be phased out by 2031

Sources: Supreme Court, M.C. Mehta v. Union of India (orders of 28 Jul 1998, 5 Apr 2002); MoEF S.O. 93(E), 29 Jan 1998; CSE (2011); CAQM Act, 2021; Rajya Sabha reply (Aug 2025).

What CNG achieved — and where it falls short

Benefits

Cleaner air from buses: replacing diesel buses with CNG helped lower PM10, carbon monoxide and SO2 levels

No fuel vapour: CNG systems are sealed, so there are no evaporative emissions

Fast, large-scale change: most of Delhi's public transport moved to CNG within about two years of April 2001

Health: less exposure to diesel exhaust, which WHO's cancer agency (IARC) lists as a known carcinogen

Limitations

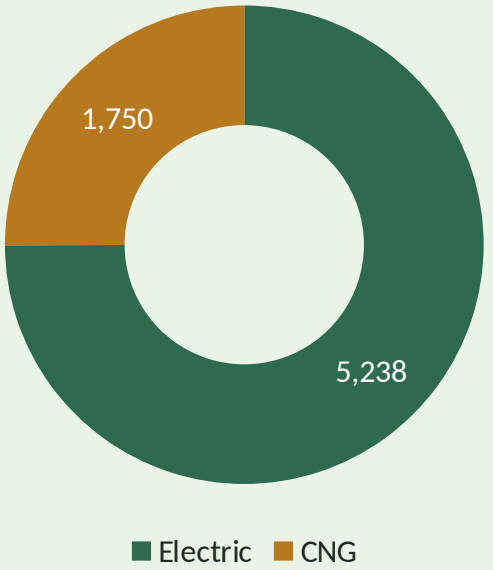
Still a fossil fuel: burning gas releases CO2, and leaked methane is a strong greenhouse gas

Gains eroded: more vehicles and more diesel cars cancelled part of the benefit

Mixed evidence: one study found NOx levels rose after the switch

Price risk: Delhi CNG rose from ₹77.09/kg (Jan 2026) to ₹86.98/kg (29 Aug 2026) as imported LNG costs climbed

Delhi's bus fleet, Oct 2026



CNG is now a transition fuel: Delhi's newest buses are electric, and its 1,750 CNG buses are due to retire.

Sources: Narain & Krupnick (2007); Ravindra et al. (2006); US DOE AFDC; IARC (2012); IGL prices via AIR News (29 Aug 2026), Deccan Herald; fleet as reported 13 Oct 2026.

Swachh Bharat Abhiyan at a glance

Launched 2 October 2014. Goal: an open-defecation-free (ODF) India by 2 October 2019, Mahatma Gandhi's 150th birth anniversary.

SBM-Grameen (rural)
Ministry of Jal Shakti

- Phase I: household toilets; villages declared ODF
- Phase II (2020–26): ODF Plus — solid & liquid waste management, visible cleanliness

SBM-Urban
Ministry of Housing & Urban Affairs

- Toilets and door-to-door waste collection in towns
- SBM-U 2.0 (from 1 Oct 2021): "Garbage Free Cities" — segregation, 3Rs, clearing old dumpsites

Government
funds, targets, monitoring

Local bodies
collection, processing, toilets

Citizens
use toilets, segregate waste

Sources: DDWS, Ministry of Jal Shakti, SBM(G) Year End Review (PIB, Jan 2026); MoHUA on SBM-U 2.0 (PIB).



Launch of the cleanliness drive, Valmiki Basti, New Delhi, 2 Oct 2014. Photo: PMO, GODL-India

Progress so far — and what remains

Official figures

12 crore+

household toilets built in rural India since 2014 (Dec 2025)

83%+

of villages declared ODF Plus (Model): about 4.89 lakh (Dec 2025)

80%

of urban solid waste processed, out of 1.62 lakh tonnes a day (Feb 2025)

Independent checks

NSS 76th round (Jul–Dec 2018): 71.3% of rural households had access to a latrine — a survey taken before the 2019 ODF declaration. Access does not always mean use.

WHO (2018): modelled 300,000+ deaths avoided (2014–19) if full coverage was reached — an estimate, not a count.

WHO/UNICEF JMP: 676 million Indians stopped open defecation over two decades (as cited by the government).

Declared vs verified: 4.89 lakh villages declared ODF Plus, 4.15 lakh verified (Dec 2025).



Ghazipur landfill, 2013. Photo: FacetsOfNonStickPans, CC BY-SA 4.0

Delhi's challenge: Ghazipur still held about 67–69 lakh tonnes of waste in 2026; the clearance target is December 2027.

Still to solve: segregation at source, processing capacity, toilet upkeep, and keeping people involved.

Sources: SBM(G) IMIS via PIB (Jan 2026); Rajya Sabha reply, MoHUA (10 Feb 2025); NSO, NSS 76th round (2019); WHO India (27 Jul 2018); MCD affidavit to NGT, OA 481/2024 and 2026 reports.

Two different policies, one shared goal

	CNG transition in Delhi	Swachh Bharat Abhiyan
Led by	Supreme Court, through a PIL (M.C. Mehta case)	Central Government with states and local bodies
Main tools	Fuel mandate, deadlines and fines	Funding, targets, campaigns, city rankings
Targets	Tailpipe emissions from public transport	Open defecation, solid and liquid waste
Economic idea	Correcting a negative externality (air pollution)	Providing public goods (sanitation, clean spaces)
Main limit	Still a fossil fuel; traffic keeps growing	Keeping toilets used and waste segregated

Sources: IIT Kanpur (2016); MoEFCC, Swachh Vayu Sarvekshan guidelines (PIB, 24 Sep 2022) and awards (PIB, 9 Sep 2025); slides 3–6.

Where they meet

- Burning waste and landfill fires pollute the air: waste burning was about 8% of winter PM2.5.
- The national clean-air ranking (Swachh Vayu Sarvekshan, NCAP) scores cities on both waste management and vehicle emissions.
- Both need strong institutions and daily citizen action.

Note: CNG is not part of Swachh Bharat Abhiyan. They are separate initiatives with overlapping health goals.

Conclusion and key references

1. Delhi's CNG switch shows a court-led fuel mandate can cut harmful pollutants quickly, but CNG is a transition fuel, not a final answer.
2. Swachh Bharat changed sanitation at national scale; official claims and independent surveys should be read together.
3. Clean air needs both cleaner vehicles and an end to open dumping and burning of waste.
4. Lasting results depend on institutions and on everyday citizen behaviour.

Full citations with links are in the speaker notes of each slide.

1. Supreme Court of India. M.C. Mehta v. Union of India, WP(C) 13029/1985 (orders: 28 Jul 1998; 5 Apr 2002).
2. MoEF (1998). Notification S.O. 93(E), 29 Jan 1998 (EPCA).
3. Narain, U. & Krupnick, A. (2007). The Impact of Delhi's CNG Program on Air Quality. RFF DP 07-06.
4. US DOE AFDC. Natural Gas Vehicle Emissions. afdc.energy.gov
5. IIT Kanpur (2016). Comprehensive Study on Air Pollution and GHGs in Delhi.
6. DDWS, Ministry of Jal Shakti (2026). SBM(G) Year End Review 2025. PIB.
7. MoHUA. SBM-Urban 2.0 (PIB); Rajya Sabha Unstarred Q. 698, 10 Feb 2025.
8. NSO (2019). NSS 76th Round: Drinking Water, Sanitation, Hygiene and Housing Condition.
9. WHO India (2018). Health gains from the Swachh Bharat initiative.
10. MoEFCC (2022, 2025). Swachh Vayu Sarvekshan. PIB.